

SAFETY DATA SHEET (GHS)

Sulfuric Acid (98% concentrate) | CAS: 7664-93-9 | Formula: H₂SO₄

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ECHA Classification & Labelling Inventory — Sulfuric acid, CAS 7664-93-9 (<https://echa.europa.eu/>)
- PubChem CID 1118 — GHS Classification (Rev.9), <https://pubchem.ncbi.nlm.nih.gov/compound/1118>
- OSHA Annotated PEL Table Z-1 — Sulfuric acid (<https://www.osha.gov/annotated-pels/table-z-1>)
- ACGIH TLV documentation — Sulfuric acid (thoracic fraction)
- Supplier SDS for Sulfuric Acid 93-98% w/w (multiple commercial sources)
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: Sulfuric Acid, 98% concentrate

CAS Number: 7664-93-9

Chemical formula: H₂SO₄

Molecular weight: 98.08 g/mol

Synonyms: Oil of vitriol; Hydrogen sulfate; Battery acid (dilute grades); Sulphuric acid

Intended use: Decomposition catalyst — CDN decomposition section, acid injection tanks D-2310/D-2311 and metering pumps P-2305A-F. PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification: Skin Corr. 1A — H314 (also covers serious eye damage)

Signal word: Danger

Hazard statements (H-codes): H314 (Causes severe skin burns and eye damage)

Precautionary statements (P-codes): P260, P264, P280, P301+P330+P331, P303+P361+P353, P304+P340, P305+P351+P338, P310, P321, P363, P405, P501

GHS pictograms: GHS05 (Corrosion)

Other hazards: Strong dehydrating agent and oxidiser; reacts violently and exothermically with water (ALWAYS add acid to water, never the reverse) and with many organic materials and bases. Inhalation of sulfuric acid MIST (as opposed to the bulk liquid) is classified by IARC as Group 1 (carcinogenic to humans) for 'strong inorganic acid mists containing sulfuric acid' — relevant to vent/atmospheric release scenarios at acid injection points. Not PBT/vPvB (inorganic substance).

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Sulfuric acid: CAS 7664-93-9 — 98% w/w

Water: CAS 7732-18-5 — 2% w/w (balance)

Note: Confirm exact concentration specification for D-2310/D-2311 acid injection grade against PPCL chemical purchase specification — concentration affects freezing point, density and corrosivity toward materials of construction

SECTION 4 — FIRST AID MEASURES

Inhalation: Remove to fresh air immediately. If breathing is difficult, administer oxygen. Obtain immediate medical attention — mist/vapour inhalation can cause severe respiratory tract burns and pulmonary oedema.

Skin contact: EMERGENCY — immediately flush with large amounts of water for at least 15-20 minutes while removing contaminated clothing (clothing may stick to skin — do not force removal if adhered, cut around it). Continue flushing en route to medical care. Obtain immediate medical attention.

Eye contact: Flush immediately with copious water for at least 15-20 minutes, holding eyelids open. Obtain immediate ophthalmological attention — risk of permanent eye damage/blindness.

Ingestion: Do NOT induce vomiting. Do NOT give anything by mouth if the person is unconscious or having convulsions. If conscious, rinse mouth thoroughly with water (do not swallow rinse water in large quantities). Obtain immediate medical attention — severe internal corrosive injury.

Notes to physician: Treat as severe chemical burn; monitor for airway compromise (inhalation), oesophageal/gastric perforation (ingestion), and acid-base (metabolic acidosis) disturbances. No specific antidote — supportive care.

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Not combustible — use extinguishing media appropriate to surrounding fire (water spray/fog from a safe distance to cool containers, dry chemical, CO₂, foam)

Unsuitable extinguishing media: Avoid direct application of large amounts of water onto concentrated acid where it can be avoided — dilution is highly exothermic and can cause violent spattering/boiling; if water must be used, apply as fine spray/fog from a safe distance

Special hazards: Non-flammable but reacts with many metals (especially when diluted or hot) to liberate flammable/explosive HYDROGEN gas, which can accumulate in confined spaces. Reaction with organic materials can be exothermic enough to ignite combustibles. Heated acid releases toxic SO₃/SO₂ fumes.

Special protective equipment for firefighters: Full acid-resistant chemical protective clothing with SCBA; approach from upwind; be alert to hydrogen gas accumulation near reacting metal surfaces

Hazardous combustion products: Sulfur oxides (SO₂, SO₃) fumes if heated; hydrogen gas if in contact with reactive metals

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Evacuate non-essential personnel. Wear full acid-resistant PPE (Section 8) with SCBA for vapour/mist releases. Eliminate sources of water contact with large spills unless deliberately diluting under controlled conditions.

Environmental precautions: Prevent entry to drains, sewers and watercourses — highly toxic to aquatic life via pH effects; neutralise before any controlled discharge per site effluent permit.

Containment and cleanup: Contain spill with acid-resistant dike/absorbent (e.g. dry sand, sodium bicarbonate, commercial acid neutralising absorbent) — do NOT use combustible absorbents. Neutralise cautiously with appropriate alkaline material (controlled addition — neutralisation is highly exothermic). Flush residual area with large amounts of water only after bulk has been neutralised/removed.

Reference to other sections: See Section 8 (PPE), Section 10 (Reactivity — water/metal reactions) and Section 13 (Disposal)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Use closed transfer systems and metering pumps (P-2305A-F) of acid-compatible construction (e.g. alloy 20, PTFE-lined, or high-silicon iron for concentrated acid). Avoid splashing; provide local exhaust ventilation at any open transfer/sampling points. Never add water to concentrated acid.

Storage conditions: Store in dedicated tanks (D-2310/D-2311) constructed of acid-resistant materials (e.g. carbon steel is acceptable for cold/concentrated 98% acid above ~80% concentration due to passivation, but becomes highly corrosive if diluted by water ingress — confirm metallurgy with corrosion engineering).

Temperature limits: Ambient storage typical; 98% sulfuric acid has a relatively high freezing point (~+3°C) — cold climates require heating/insulation to prevent freezing and associated tank stress; avoid uncontrolled heating which increases corrosivity and fume generation

Pressure: Atmospheric storage with vent to safe location (acid fume scrubber recommended); avoid pressurisation

Segregation requirements: Segregate from all bases/alkalis, water (bulk contact), organic materials, oxidisable materials, metals (especially active metals), cyanides, sulfides and chlorates/nitrates (violent reactions)

Special requirements: Bunded secondary containment for acid storage and injection skid; dedicated eyewash/safety shower immediately adjacent to D-2310/D-2311 and P-2305A-F; leak detection on metering pump seals given the critical, continuous-dosing role of acid catalyst in the decomposer

Specific end-use recommendations: Maintain metering accuracy — both under- and over-dosing of acid catalyst to the CDN decomposer are process safety concerns (incomplete CHP decomposition vs. accelerated/uncontrolled decomposition)

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): ACGIH TLV-TWA: 0.2 mg/m³ (thoracic particulate mass), with an A2 (suspected human carcinogen) notation applicable to strong inorganic acid mists containing sulfuric acid. OSHA PEL: 1 mg/m³ TWA. Thailand OEL: confirm current value with Ministry of Labour notification (commonly aligned to 1 mg/m³ TWA).

TWA: 0.2 mg/m³ (ACGIH, thoracic fraction) / 1 mg/m³ (OSHA)

STEL: Not formally established by ACGIH; some jurisdictions apply a STEL — confirm locally

Ceiling: Not established

Skin notation: Not formally assigned (corrosive — direct contact hazard addressed under Skin Corr. 1A)

IARC/NTP/OSHA carcinogen status: IARC Group 1 — carcinogenic to humans (specifically for occupational exposure to strong inorganic acid mists containing sulfuric acid, e.g. from pickling/electroplating-type mist-generating operations)

Engineering controls: Closed handling systems; local exhaust ventilation/mist scrubbing at any vent or open transfer point; continuous leak/level monitoring on acid tanks and metering pumps

PPE — Respiratory: Full-facepiece respirator with acid-gas cartridge for tasks with mist potential; SCBA for emergency response/major release

PPE — Gloves: Acid-resistant gloves (e.g. butyl rubber, neoprene, PVC, or laminate film) rated for concentrated sulfuric acid — confirm breakthrough time

PPE — Eyes: Chemical splash goggles plus full face shield for any connection/sampling task

PPE — Body: Acid-resistant apron/suit (e.g. PVC- or neoprene-coated); full acid suit for tank entry/major spill response

Biological monitoring: Not applicable/established

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Colourless to slightly yellow/brown oily liquid

Odour: Odourless (bulk liquid); pungent/acrid if fuming or generating SO₃ mist

Odour threshold: Not established (relevant hazard is mist/acidity, not odour perception)

pH: Strongly acidic — pH <1 even when significantly diluted

Melting point / Freezing point: ~+3°C for 98% w/w (freezing point varies with concentration — minimum freezing point for sulfuric acid/water mixtures occurs around 93-94% w/w at approx. -35°C; the 98% grade freezes at a comparatively higher temperature, relevant to cold-weather tank heating design)

Boiling point (at 1 atm): ~290-338°C (98% w/w); decomposes releasing SO₃ before/at boiling

Flash point: Not applicable — non-flammable

Evaporation rate: Negligible (very low vapour pressure)

Flammability (solid/gas): Not flammable; can support fire indirectly via reaction with combustibles/metals (hydrogen generation)

Flammability limits in air: Not applicable

Vapour pressure (at 20°C): <0.001 kPa (negligible)

Vapour density (air = 1): ~3.4 (for H₂SO₄ vapour/SO₃, where present — relevant to fume behaviour)

Relative density (water = 1): ~1.84 (98% w/w at 20°C)

Solubility (water): Fully miscible — dissolution is strongly exothermic

Partition coefficient (log K_{ow}): Not applicable (inorganic acid; fully dissociates in water)

Auto-ignition temperature: Not applicable

Decomposition temperature: Begins releasing SO₃ (fuming) on heating well below boiling point; full thermal decomposition to SO₂/SO₃/H₂O/O₂ occurs at high temperatures (>300°C)

Viscosity: ~25 mPa·s (98% w/w at 20°C — significantly more viscous than water)

Explosive properties: Not classified as explosive; explosion hazard arises indirectly via hydrogen generation in confined spaces from metal reactions

Oxidising properties: Acts as an oxidiser/dehydrating agent in concentrated form, particularly when hot

Molecular weight: 98.08 g/mol

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Highly reactive — strong acid, strong dehydrating agent and oxidiser when concentrated/hot

Chemical stability: Stable under normal storage conditions in compatible containment

Possibility of hazardous reactions: Violent, highly exothermic reaction with water (can boil/spatter if water is added to acid); reacts with many metals to generate flammable hydrogen gas; reacts violently with bases (neutralisation exotherm); can ignite/char organic materials on contact (dehydration); reacts violently with cyanides/sulfides to release toxic gases (HCN/H₂S) and with chlorates/perchlorates/permanganates (explosion risk)

Conditions to avoid: Addition of water to concentrated acid; contact with reactive metals in confined spaces (H₂ accumulation); contact with strong oxidisers, organics, cyanides/sulfides

Incompatible materials: Water (uncontrolled mixing), bases/alkalis, most metals (especially in dilute acid), organic materials, cyanides, sulfides, chlorates, nitrates, permanganates, carbides, peroxides

Hazardous decomposition products: Sulfur trioxide (SO₃) and sulfur dioxide (SO₂) fumes on heating/decomposition; hydrogen gas on contact with active metals

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Dermal, ocular, inhalation (mist), ingestion

Acute toxicity — Oral LD50: ~2,140 mg/kg, rat (corrosive effects dominate over systemic toxicity)

Acute toxicity — Dermal LD50: Not well characterised — corrosive effects dominate; treat as corrosive substance

Acute toxicity — Inhalation LC50: ~510 mg/m³ (2 h), rat, mist (approximate, literature range) — severe respiratory tract irritation/corrosion at much lower concentrations

Skin corrosion/irritation: Causes severe burns (H314) — full-thickness chemical burns possible on contact with concentrated acid

Serious eye damage/irritation: Causes serious eye damage — risk of permanent injury/blindness

Respiratory/skin sensitisation: Not classified as a sensitiser

Germ cell mutagenicity: Not classified

Carcinogenicity: IARC Group 1 for occupational exposure to strong inorganic acid mists containing sulfuric acid (mist inhalation, not bulk liquid contact); ACGIH A2 notation for thoracic particulate fraction

Reproductive toxicity: Not classified

STOT (single exposure): Severe respiratory tract irritation/corrosion on inhalation of mist/fume

STOT (repeated exposure): Repeated mist exposure associated with chronic respiratory effects and dental erosion; see carcinogenicity note above

Aspiration hazard: Not classified (corrosive liquid — aspiration would cause severe chemical injury but is not assessed under standard aspiration toxicity criteria)

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC50/EC50): Toxicity to aquatic organisms is primarily a function of pH depression; LC50 values are concentration/pH dependent — neutralise before any discharge

Persistence and degradability: Not applicable — inorganic acid; neutralises in the environment to sulfate salts

Bioaccumulative potential: Not applicable

Mobility in soil: Highly mobile (water miscible); can acidify soil/groundwater

Other adverse effects: Significant pH-related impacts on receiving waters and soils if released uncontrolled; no ozone-depletion potential

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Neutralise (e.g. with lime/soda ash) under controlled conditions (highly exothermic) prior to disposal as per local effluent/waste regulations; large quantities should be handled by a licensed hazardous waste contractor

Contaminated packaging: Rinse with water (controlled, with PPE) before disposal/reuse; dispose of residues as hazardous waste

Regulatory reference: Thailand Hazardous Substance Act B.E. 2535; site effluent discharge permit conditions (pH limits)

SECTION 14 — TRANSPORT INFORMATION

UN Number: UN 1830

Proper shipping name: SULFURIC ACID (with not less than 51% acid)

Transport hazard class(es): Class 8 (Corrosive)

Packing group: II

Environmental hazards (marine pollutant): Not classified as a marine pollutant under GHS aquatic criteria, but corrosive/pH hazards apply to marine environment

Special precautions for user: Transport only in compatible (acid-resistant) tanks; ensure adequate venting — do not transport in sealed containers without appropriate venting provisions

Bulk transport (MARPOL/IBC): Classify per IBC Code if shipped in bulk by sea

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 7664-93-9)

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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